CCSA105 Practical Assignment: Policy Layers

Course: CCSA105 Policy Layers (Module 5)

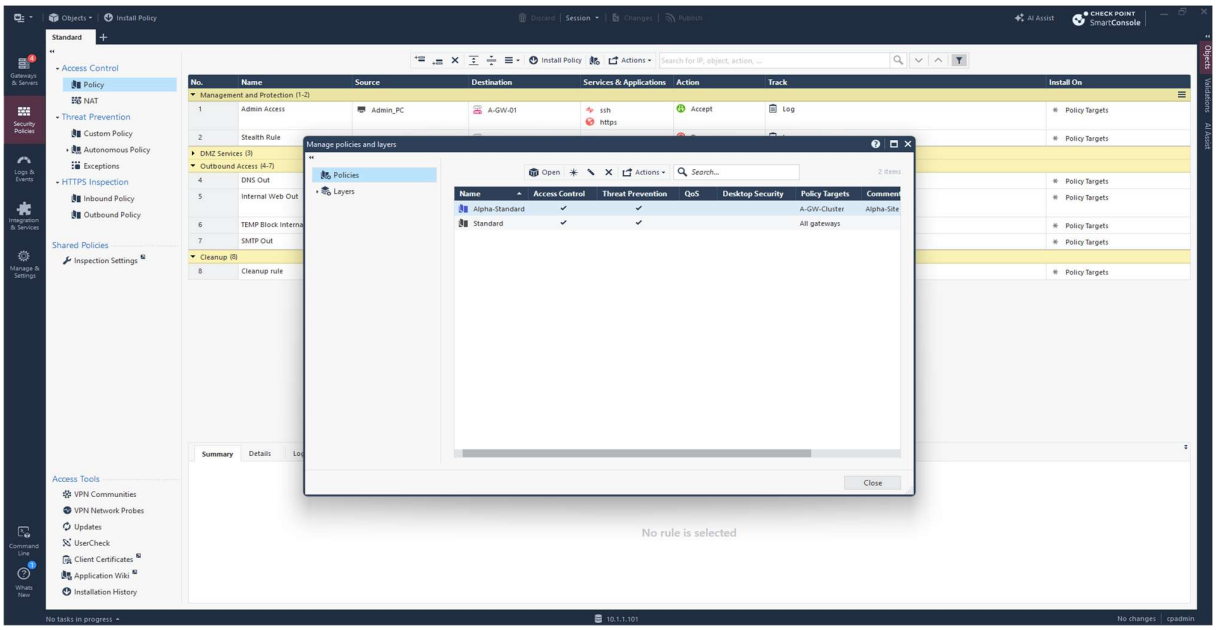
Student name	Ogagaoghene Esi	**Student ID**	C11/26/CCSA/17062
Date	11-11-2026	**Score**	/ 100

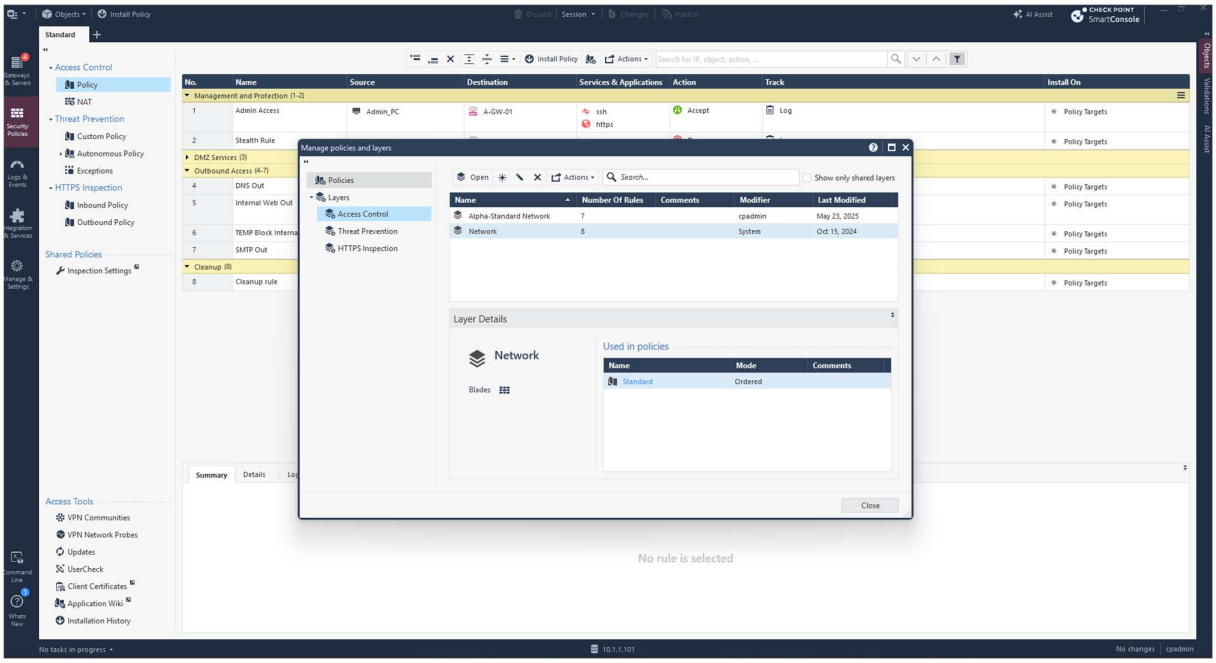
Part A: Practical Tasks

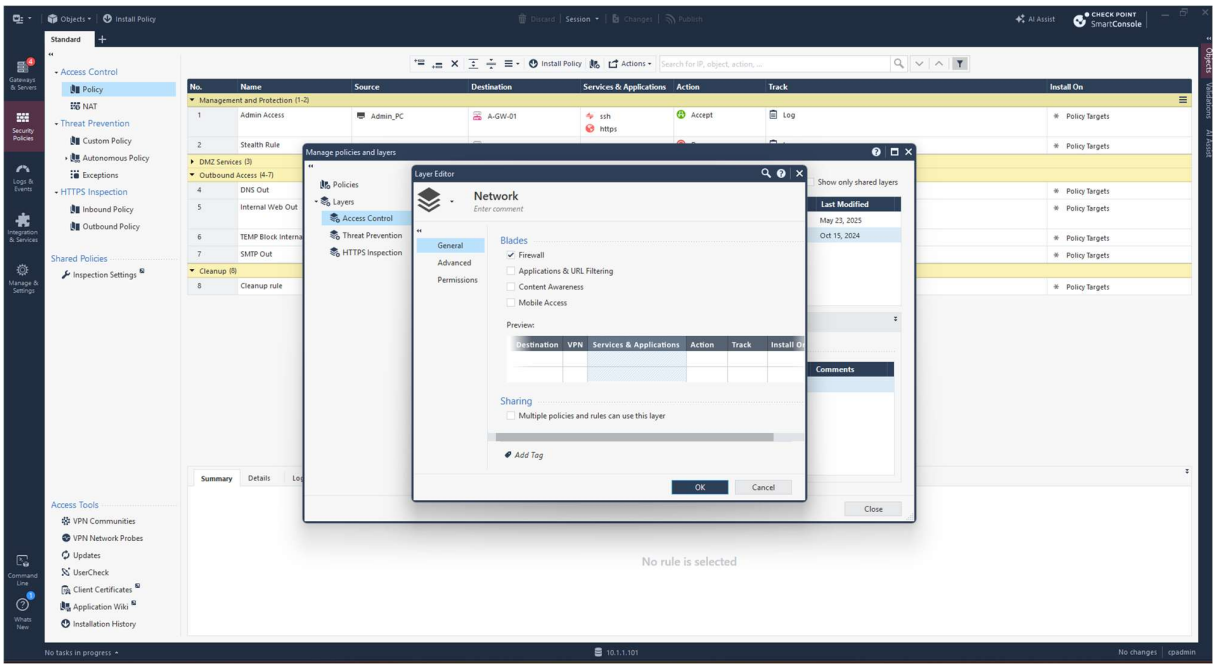
Task 1: Explore the default policy and layer

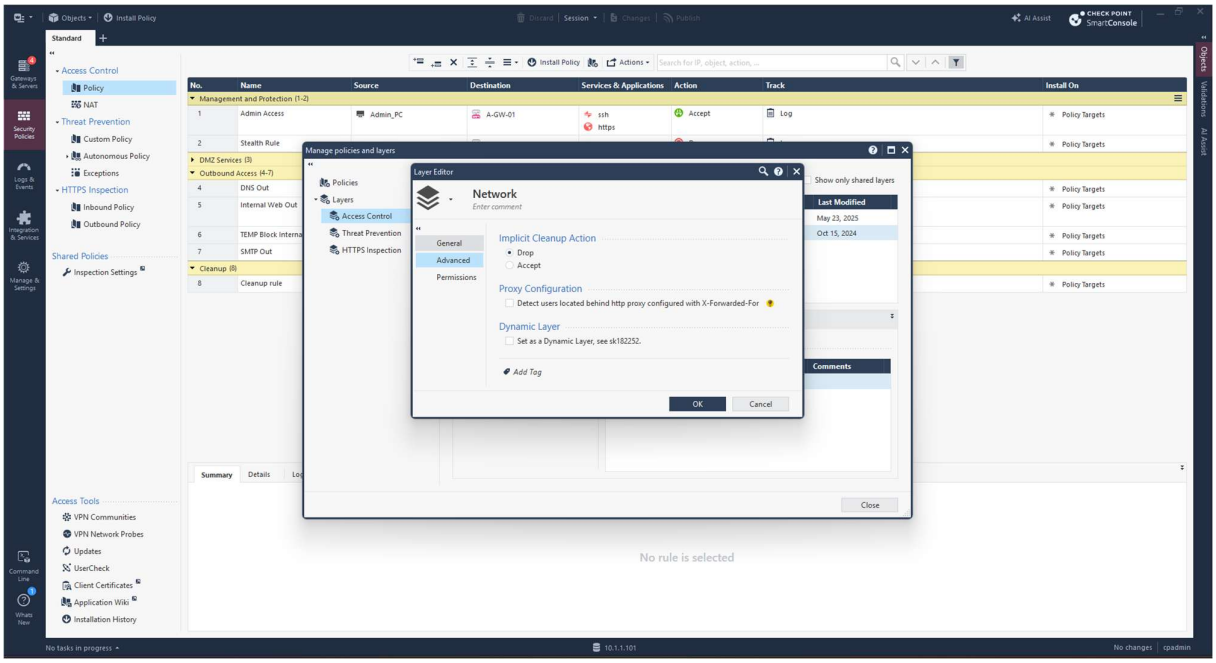
1. In SmartConsole, open the **Applications** menu and choose **Manage policies and layers**. Record the name of the default Access Control policy and the name of its default Ordered Layer.
2. Open the properties of the default Ordered Layer and record its settings on three pages: **General** (blades enabled, Sharing, which policies it applies to), **Advanced** (Implicit Cleanup Action, Proxy Configuration, Dynamic Layer) and **Permissions** (which profiles can edit it).
3. Open the default policy's rulebase, find its Cleanup rule and record its Action.

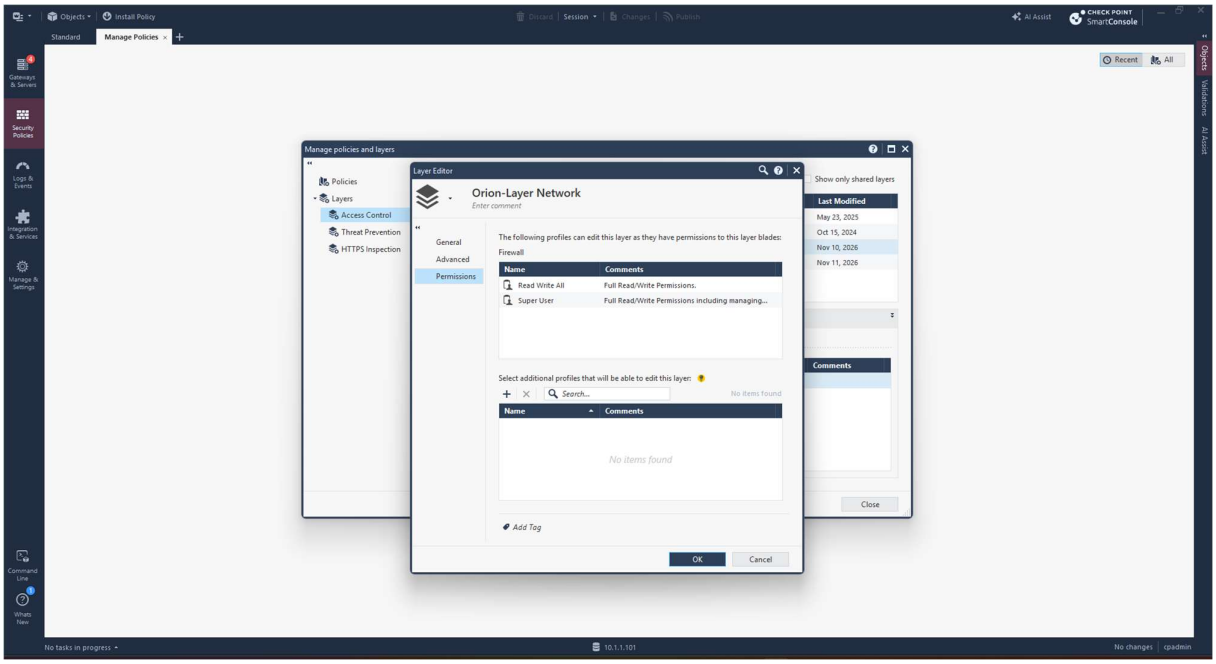
Evidence:

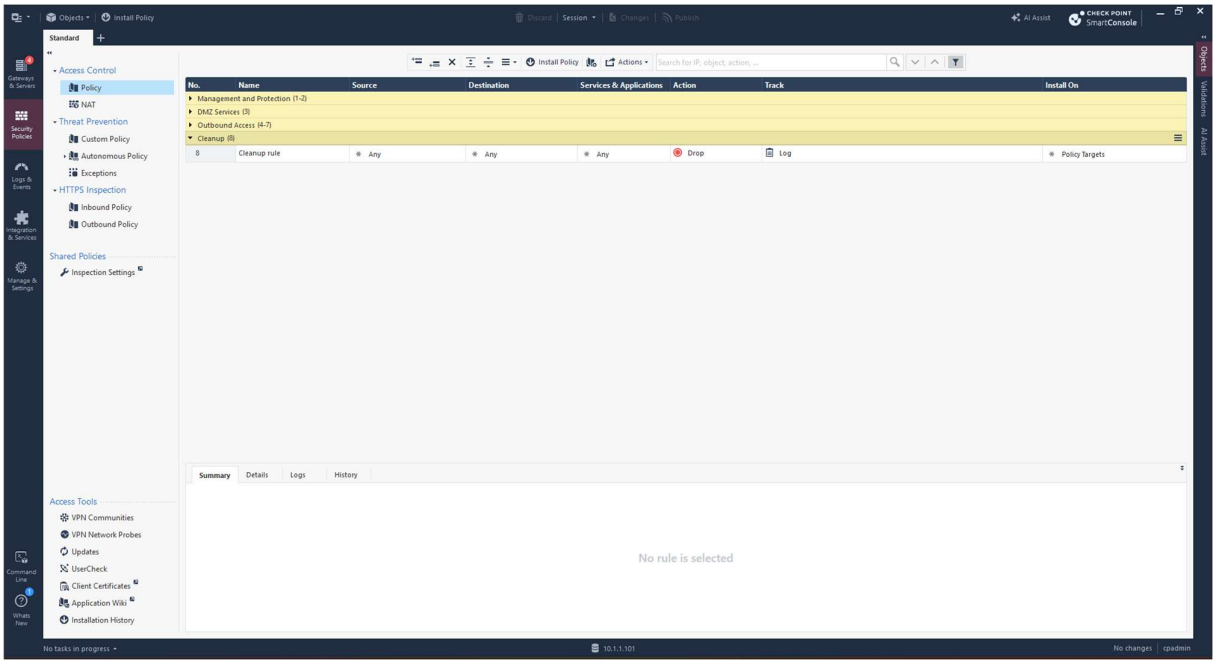



Manage policies and layers
Policies
Layers
Access Control
Threat Prevention
HTTPS Inspection
Open
Actions
Search...
Show only shared layers
Name
Number Of Rules
Comments
Modifier
Last Modified
Alpha-Standard Network
7
cpadmin
May 23, 2025
Network
8
System
Oct 15, 2024
Layer Details
Network
Blades
Used in policies
Name
Mode
Comments
Standard
Ordered
Close
No rule is selected


Manage policies and layers
Layer Editor
Network
Enter comment
General
Advanced
Permissions
Blades
Firewall
Applications & URL Filtering
Content Awareness
Mobile Access
Preview:
Destination
VPN
Services & Applications
Action
Track
Sharing
Multiple policies and rules can use this layer
Add Tag
OK
Cancel
Comments
Close
No rule is selected


Manage policies and layers
Layer Editor
Network
Enter comment
Implicit Cleanup Action
Drop
Accept
Proxy Configuration
Detect users located behind http proxy configured with X-Forwarded-For
Dynamic Layer
Set as a Dynamic Layer, see sk182252.
Add Tag
OK
Cancel
Close
No rule is selected





Task 2: Create a policy package and build the Network layer

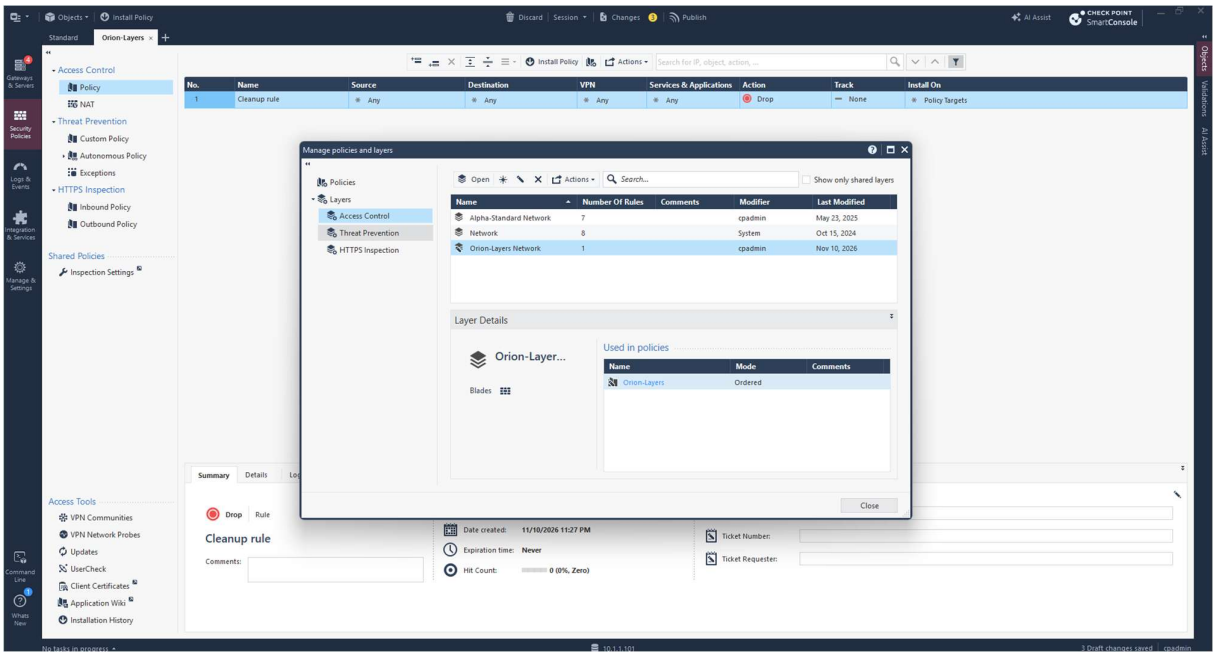
This task starts the work of Lab 5A.

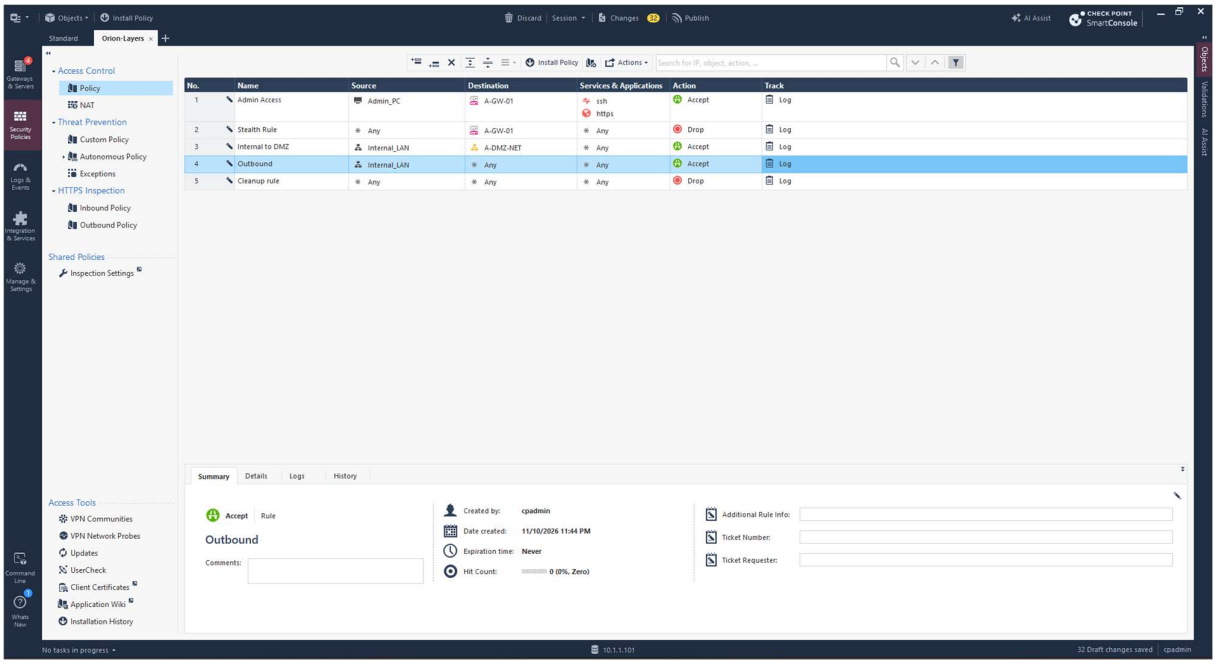
1. In **Manage policies and layers**, create a new policy package named **Orion-Layers** with Access Control enabled. Check the Ordered Layer it contains. Make sure it is named **Network**, and rename it if it is not.
2. Build the Network layer rulebase as in the table below.
3. Leave rule 3 with the Action **Accept** for now. You will change it in Task 5.

4. Add an explicit **Cleanup Rule** at the end of the layer. Its Action must be **Drop**, the same as the layer's Implicit Cleanup Action.

Position	Name	Source	Destination	Services & Applications	Action	Track
1	Admin Access	Admin_PC	Gateway	ssh, https	Accept	Log
2	Stealth Rule	Any	Gateway	Any	Drop	Log
3	Internal to DMZ	Internal_LAN	DMZ_Net	Any	Accept (changed in Task 5)	Log
4	Outbound	Internal_LAN	Any	Any	Accept	Log
5	Cleanup Rule	Any	Any	Any	Drop	Log

Evidence:



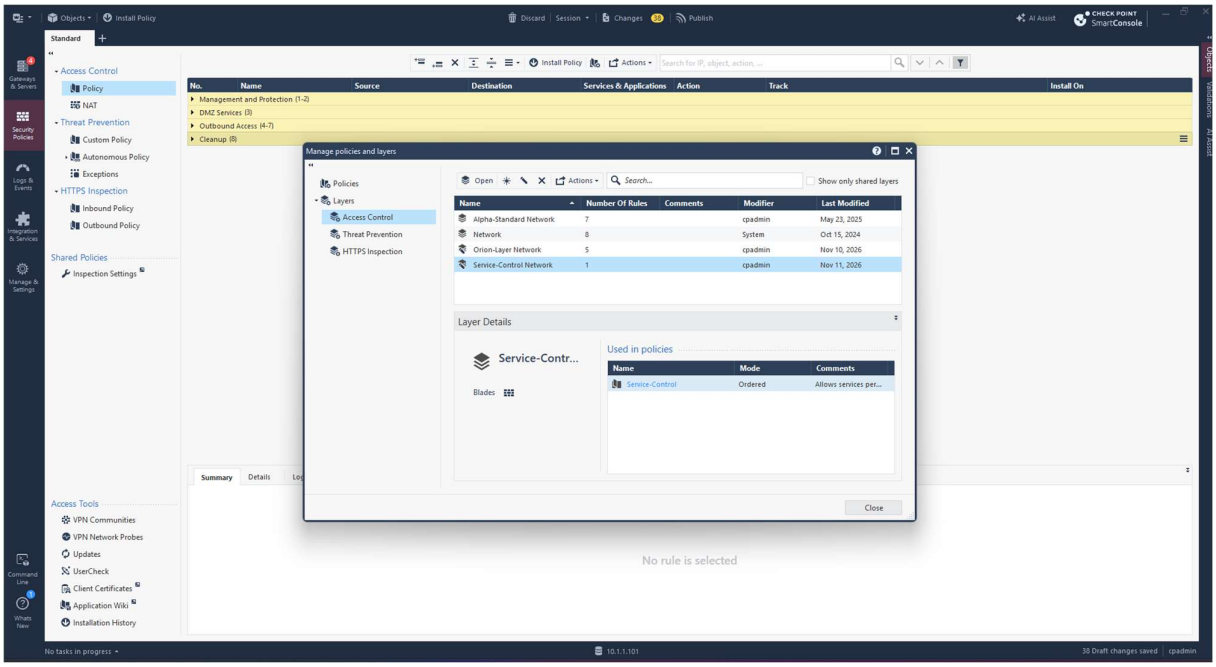


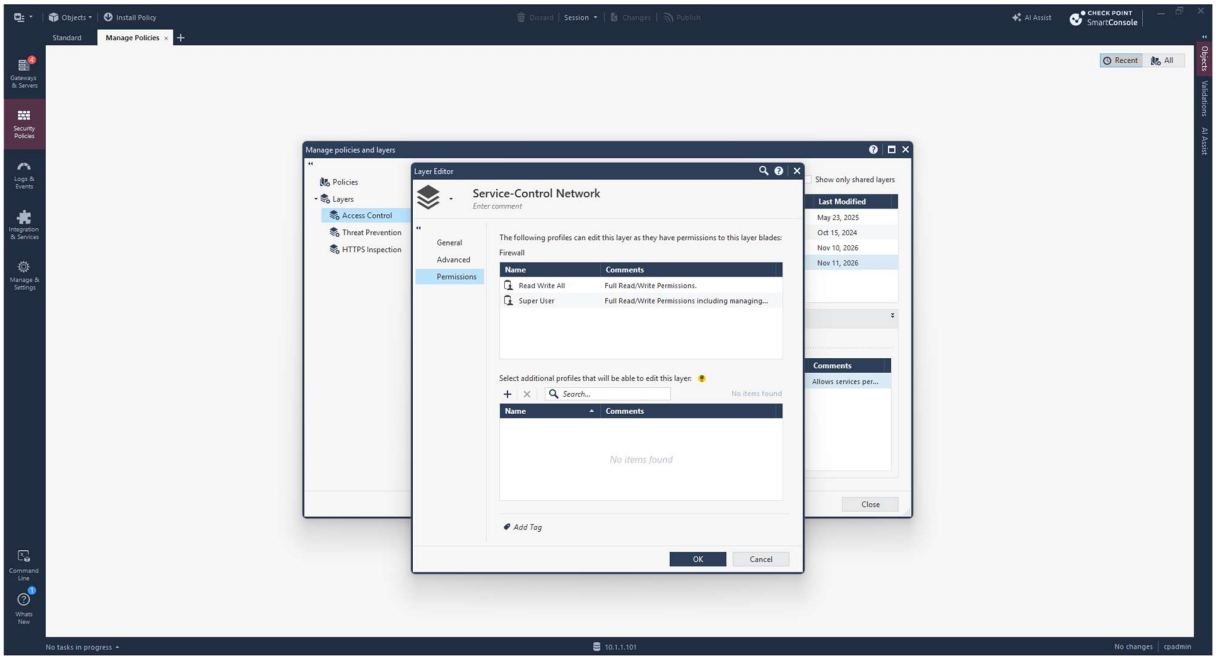
Task 3: Add a second Ordered Layer

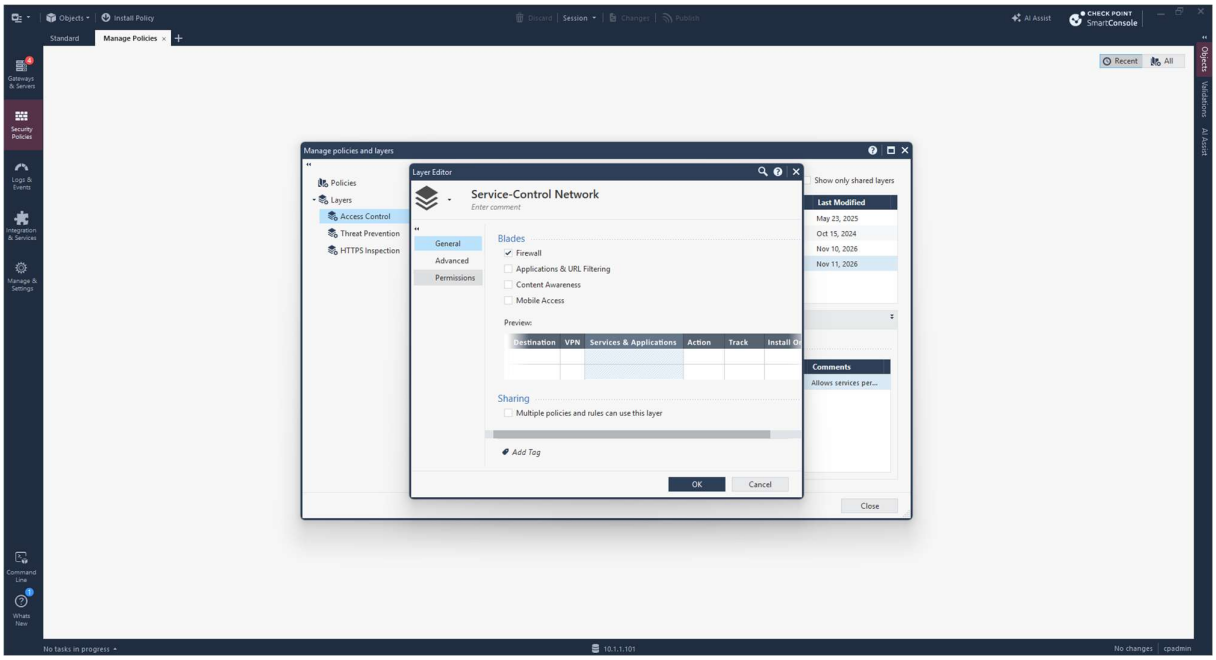
1. In **Manage policies and layers**, open the **Layer Editor** and add a new layer.
2. On the **General** page: set the Name to **Service-Control**, add a Comment, enable only the **Firewall** blade and set Sharing to **Disabled**.
3. On the **Advanced** page: set the Implicit Cleanup Action to **Drop** and leave Proxy Configuration and Dynamic Layer disabled.
4. On the **Permissions** page: leave **Super User** and **Read-Write All** able to edit the layer.
5. Add the layer to the **Orion-Layers** policy as the second Ordered Layer, **below** Network.
6. Build the Service-Control rulebase as in the table below, then publish.
7.

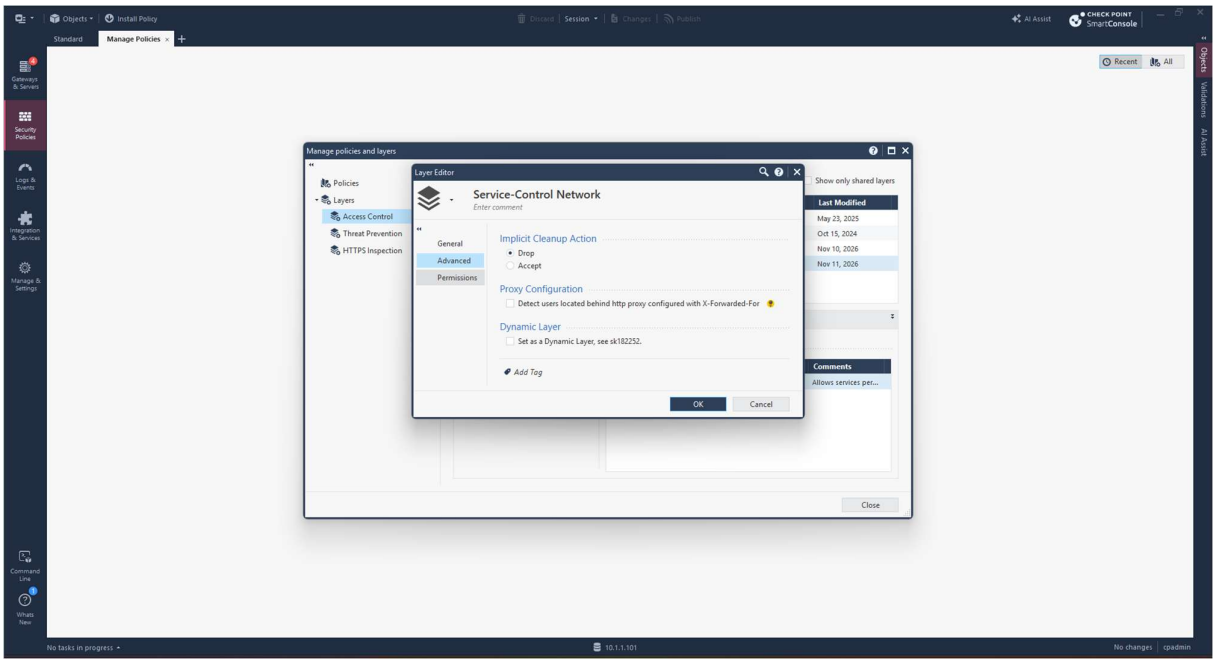
Position	Name	Source	Destination	Services & Applications	Action	Track
1	Allowed Services	Any	Any	http, https, dns (UDP and TCP), smtp	Accept	Log
2	Cleanup Rule	Any	Any	Any	Drop	Log

Evidence:


Manage policies and layers
Policies
Layers
Access Control
Threat Prevention
HTTPS Inspection
Layer Details
Service-Contr...
Used in policies
No rule is selected
Close


Manage policies and layers
Layer Editor
Service-Control Network
Enter comment
General
Advanced
Permissions
The following profiles can edit this layer as they have permissions to this layer blades:
Firewall
Read Write All
Full Read/Write Permissions.
Super User
Full Read/Write Permissions including managing...
Select additional profiles that will be able to edit this layer:
No items found
Add Tag
OK
Cancel
Close





Task 4: Install, test and fix

1. Install the **Orion-Layers** policy package on your gateway. Confirm the installation succeeds.
2. Run test **T1** only. Record the result and the log entry, noting the **layer** and **rule** that matched. The result will differ from what the Network layer alone suggests. Explain why in two sentences.
3. Fix the problem by adding a rule named **Admin Services** at position 1 of the Service-Control layer: Source Admin_PC, Destination Gateway, Services ssh and https, Action Accept, Track Log. Publish and install again.

4. Run tests **T1 to T5**. For each, record the result and the layer and rule that matched. Adapt the hosts and tools to what your lab provides.

Test	From	Action
T1	Admin_PC	Connect to the gateway using ssh
T2	Another internal host	Browse to an external server on http
T3	Another internal host	Ping an external server
T4	Another internal host	Ping the gateway's internal interface
T5	Another internal host	Connect to Web_Server on telnet (tcp 23)

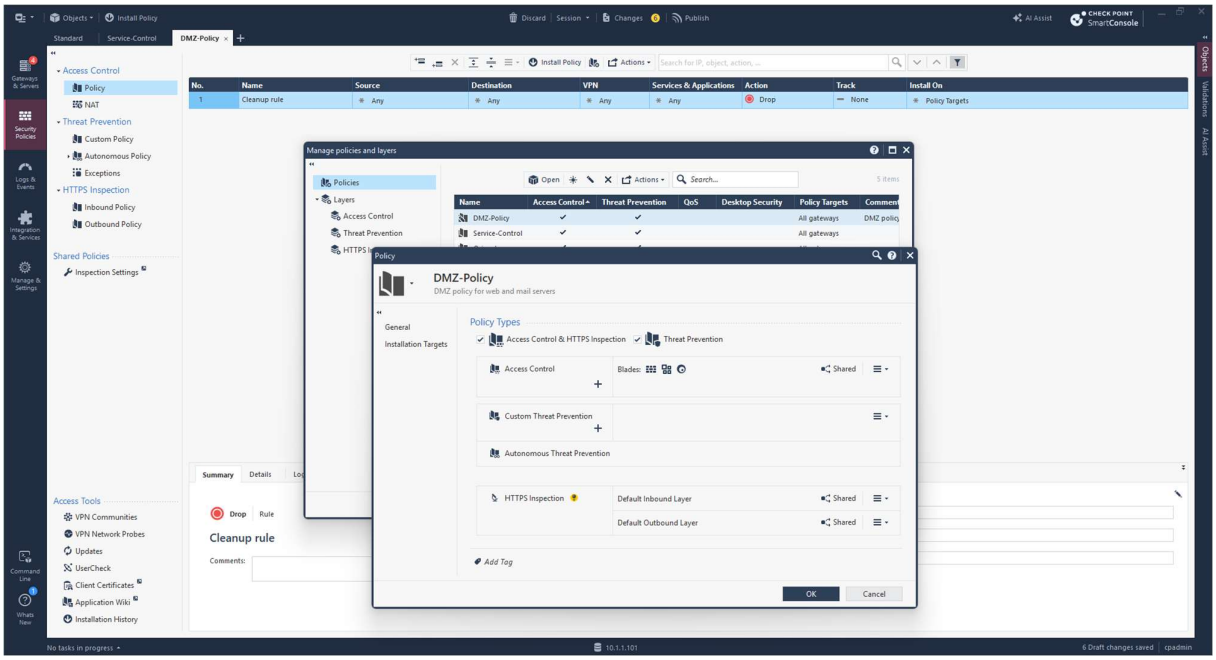
Evidence: I can't install the policy on my gateway because I don't have the VM installed. I requested for the link to enable me download and install them locally on my second laptop but I was not provided with the requested link. The cloud environment is not adequate for this practice lab, hence, I can't produce screenshots for this Task as evidence. But I understand the logic and essence of the task.

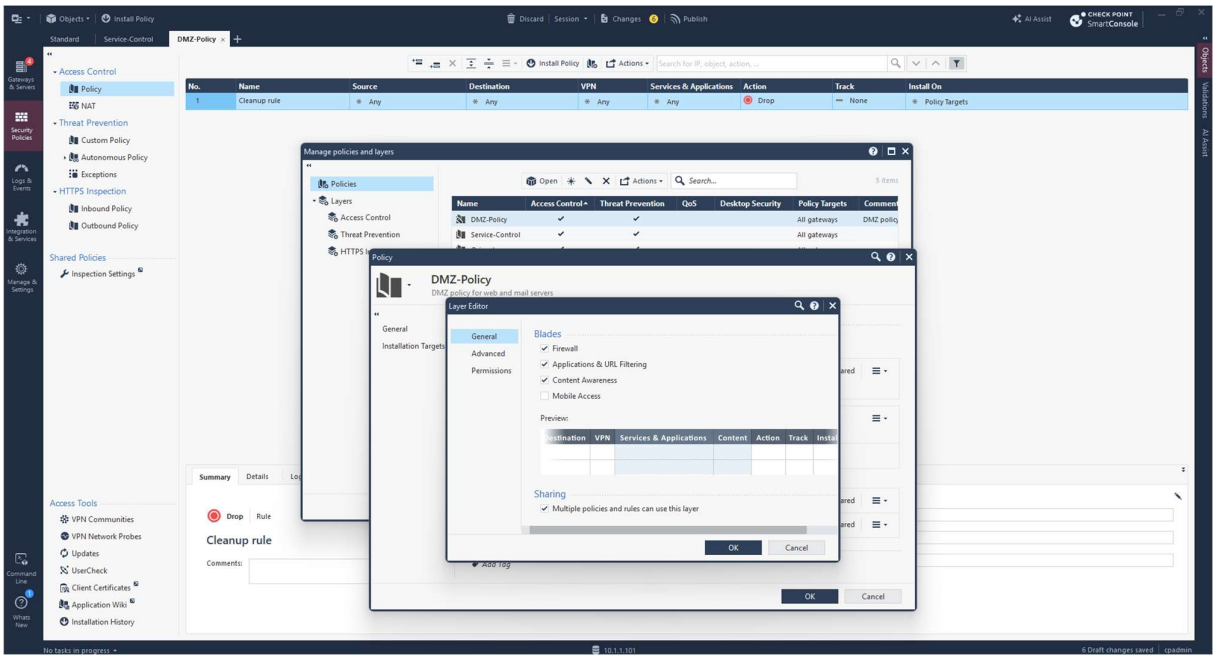
Task 5: Create an Inline Layer

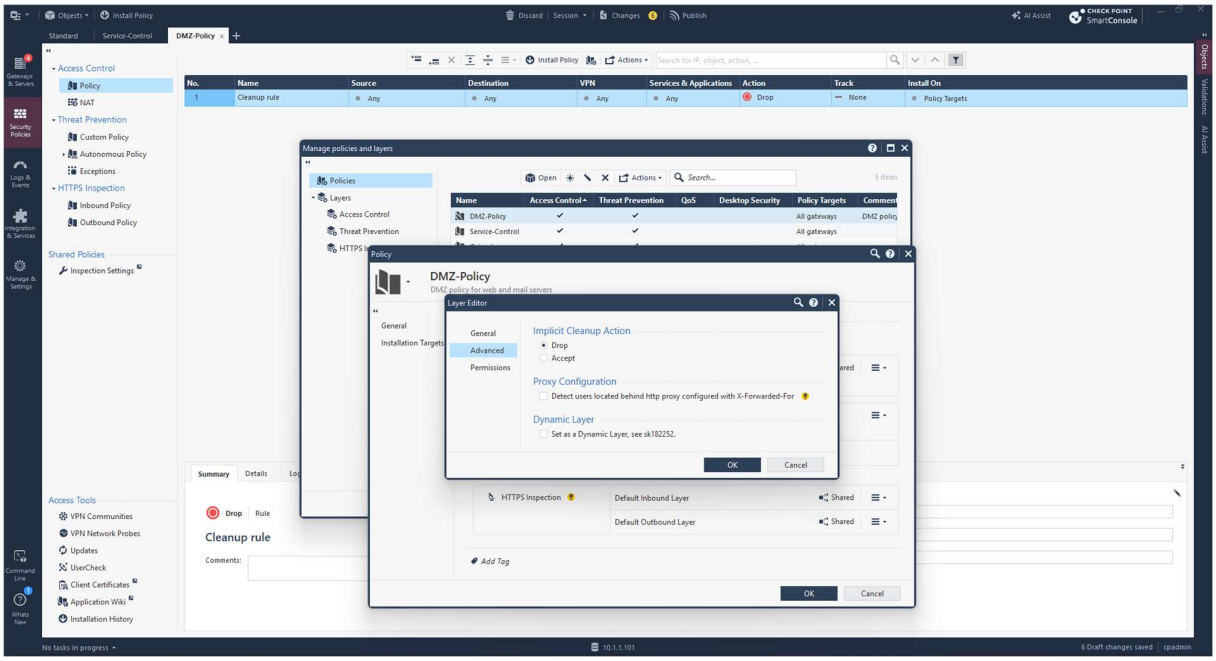
1. In the **Layer Editor**, add a new layer. On the **General** page: set the Name to **DMZ-Policy**, add a Comment, enable all blades **except Mobile Access** (Firewall, Applications & URL Filtering and Content Awareness) and set Sharing to **Enabled**. If your lab gateway does not have those blades, enable Firewall only and say so in your report.
2. On the **Advanced** page: set the Implicit Cleanup Action to **Drop**. On the **Permissions** page: leave **Super User** and **Read-Write All** able to edit the layer.
3. Build the DMZ-Policy rulebase as in the table below.
4. In the Network layer, edit rule 3 (**Internal to DMZ**). Change its Action from **Accept** to **Inline Layer** and select **DMZ-Policy**. Publish.

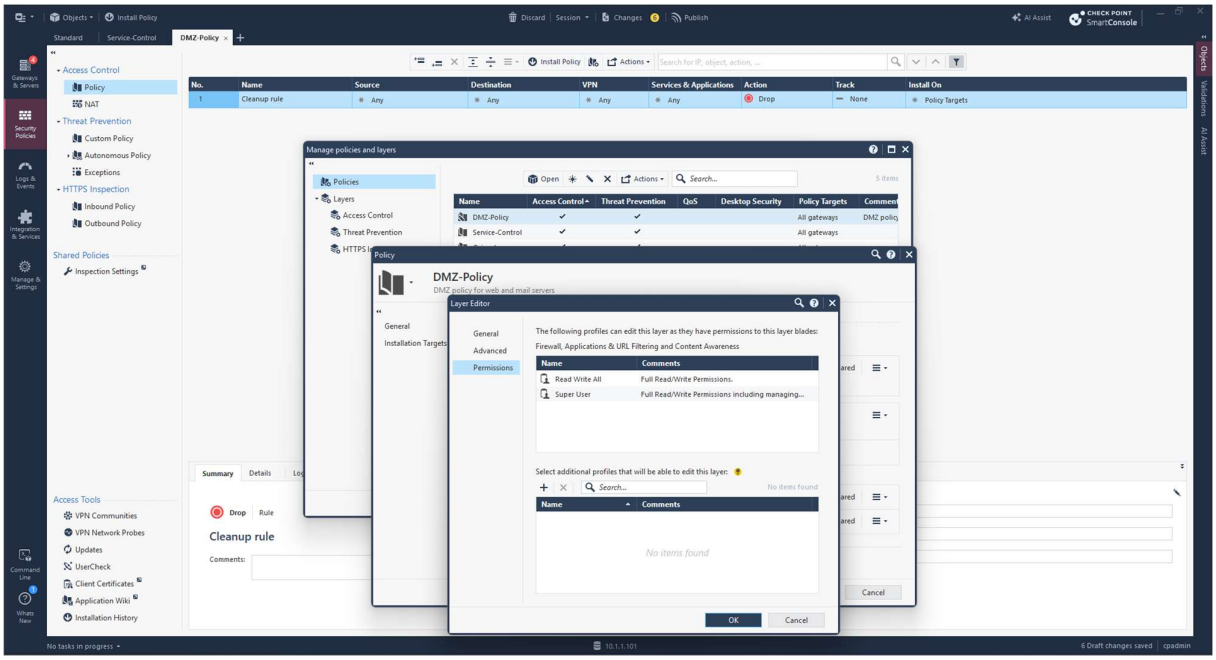
Position	Name	Source	Destination	Services & Applications	Action	Track
1	Web Access	Any	Web_Server	http, https	Accept	Log
2	Mail Access	Any	Mail_Server	smtp	Accept	Log
3	Cleanup Rule	Any	Any	Any	Drop	Log

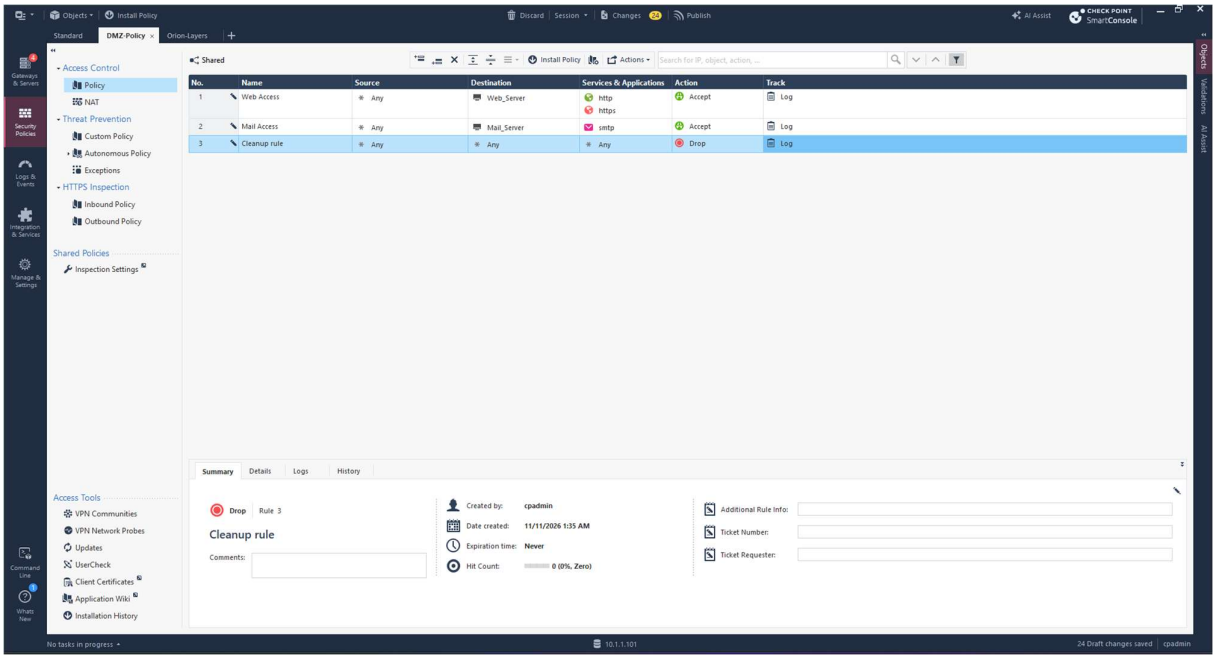
Evidence:

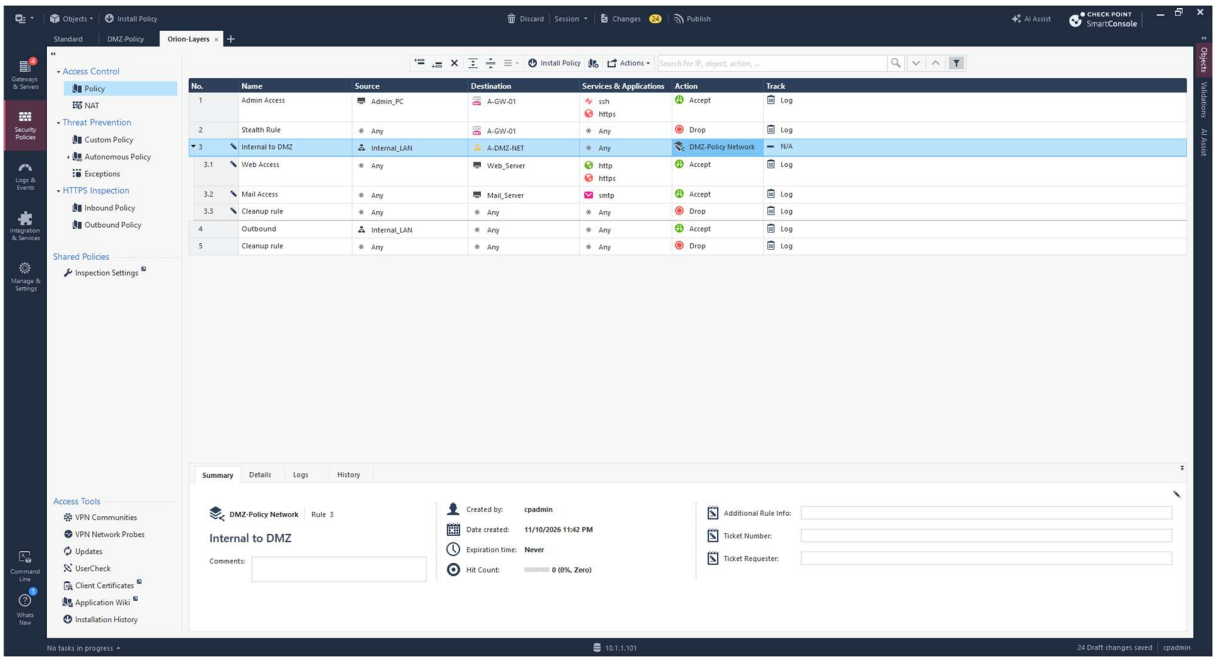

Manage policies and layers
Policy
DMZ-Policy
DMZ policy for web and mail servers
Policy Types
Access Control & HTTPS Inspection
Threat Prevention
Cleanup rule
OK
Cancel


Manage policies and layers
Policy
DMZ-Policy
DMZ policy for web and mail servers
Layer Editor
General
Advanced
Permissions
Blades
Firewall
Applications & URL Filtering
Content Awareness
Mobile Access
Preview:
Sharing
Multiple policies and rules can use this layer
OK
Cancel
Cleanup rule


DMZ-Policy
Layer Editor
Implicit Cleanup Action
Drop
Accept
Proxy Configuration
Detect users located behind http proxy configured with X-Forwarded-For
Dynamic Layer
Set as a Dynamic Layer, see sk182252.
OK
Cancel
Cleanup rule


DMZ-Policy
Layer Editor
The following profiles can edit this layer as they have permissions to this layer blades:
Firewall, Applications & URL Filtering and Content Awareness
Read Write All
Full Read/Write Permissions.
Super User
Full Read/Write Permissions including managing...
Select additional profiles that will be able to edit this layer:
No items found
OK
Cancel
Cleanup rule





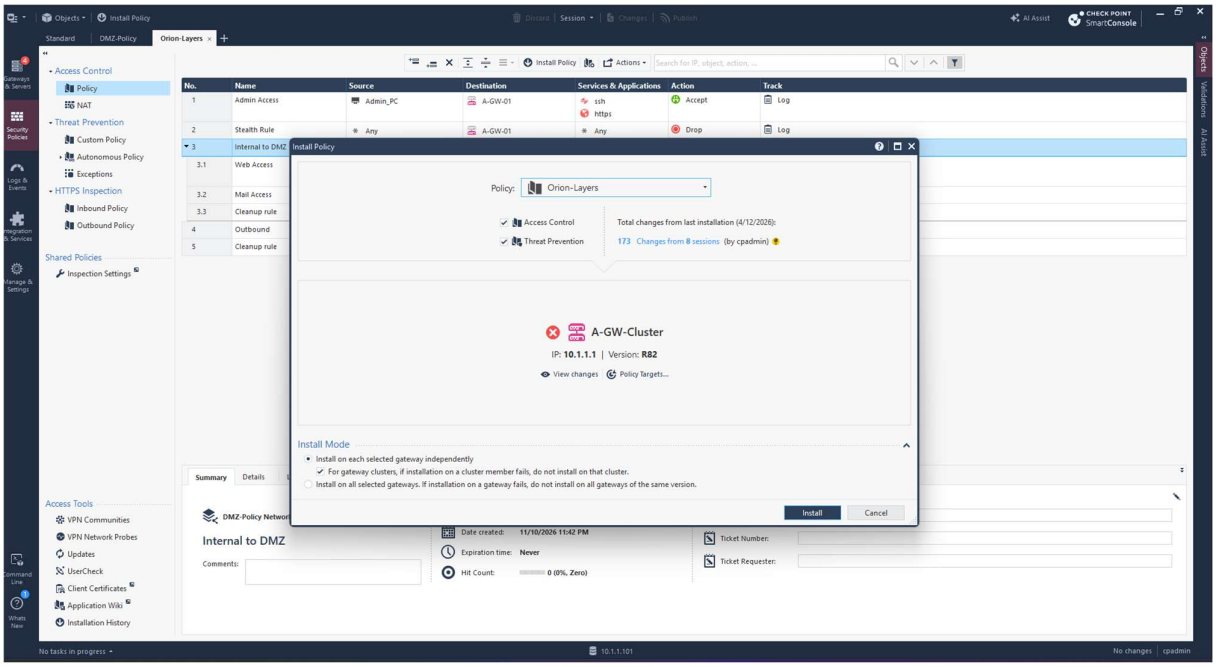
Task 6: Install and verify the Inline Layer

1. Install the **Orion-Layers** policy package and confirm that it succeeds.
2. Run tests **T6 to T9**. For each, record the result and the layer and rule that matched.

Test	From	Action
T6	Another internal host	Browse to Web_Server on http
T7	Another internal host	Connect to Mail_Server on smtp (tcp 25)
T8	Another internal host	Connect to Web_Server on telnet (tcp 23). This repeats T5.
T9	Another internal host	Browse to an external server on http

3. Compare the T5 and T8 logs. Explain which layer dropped the packet each time, and why the answer changed after you added the Inline Layer.

Evidence:



Task 7: Layer order and sharing

1. Move **Service-Control** above **Network** in the policy's layer list (for example, right-click the Access Control policy and choose Edit Policy, or follow your lab guide). Publish and install.
2. Repeat tests **T3** and **T4**. Record the layer and rule that matched each time.
3. Explain why the packets are dropped by a different layer than before, and why the outcome for the user is the same.
4. Move the layers back to the original order, then publish and install.
5. Open the **DMZ-Policy** layer in the Layer Editor. Explain how its **Sharing: Enabled** setting would let you reuse it in another policy package.

Evidence:

Part B: Theory Questions

Section B1

Q1: B. A layer is a manageable collection of access rules within a policy.

Q2: B — Standard and Firewall. This is the course-expected default naming; confirm actual names in Task 1, since installations can be customized.

Q3: A. An Accept advances to the next Ordered Layer; all applicable layers must allow.

Q4: D. An Inline Layer is a sub-policy called from a specific parent rule.

Q5: B. An explicit Cleanup Rule makes the final action visible and loggable and should match implicit cleanup.

Section B2

Q6. A policy layer is a separately manageable rulebase within an Access Control policy. It helps a large enterprise by dividing a long rulebase into smaller logical responsibilities. Two benefits are easier review/troubleshooting and more focused administrator permissions.

Q7. Ordered Layers are evaluated sequentially for traffic, and an Accept must continue to subsequent Ordered Layers; for example, Network and Service-Control separate network reachability from permitted protocols. An Inline Layer is evaluated only when traffic matches its parent rule; for example, Internal to DMZ invokes DMZ-Policy to limit access to Web_Server and Mail_Server.

Q8. The gateway checks Ordered Layers in their configured order. A Drop in any layer rejects traffic immediately; Accept continues to the next applicable layer, and traffic is allowed only if it survives all required layers. If no explicit rule matches, the layer's implicit cleanup action applies (Drop in this assignment); an explicit Cleanup Rule makes that decision visible in the rulebase and logs.

Q9. General: enabled blades or Sharing; Advanced: Implicit Cleanup Action or Dynamic Layer; Permissions: profiles such as Super User that may edit the layer.

Q10. Sharing controls whether a layer can be reused by multiple policy packages. Enable it for deliberately reusable, centrally maintained rulebases, but check the impact of edits on every package that references the layer and apply appropriate permissions.

Section B3

Q11. The connection is dropped: the first Ordered Layer's Accept does not override the second layer's implicit Drop when no rule matches. I would check Logs & Monitor. If it should be allowed, I would add a narrow Accept rule for that traffic in the second layer, publish, install and retest. An explicit Cleanup Rule with Drop/Log is recommended because it makes the fallback action easy to audit and troubleshoot.

Q12. I would use an Ordered Network layer for topology and administrative access, followed by an Ordered Service-Control layer for protocol restrictions. Specific high-risk branches such as DMZ access would call Inline Layers for web/mail services, with explicit Cleanup Drop/Log at each layer. This reduces the number of rules administrators need to inspect at once, supports separate edit permissions for specialized teams, and makes policy ownership and review clearer. Performance is not automatically faster merely because layers exist: inspection can involve additional rules and enabled security blades, so any performance claim should be validated through gateway monitoring and traffic tests